

Benton Tripp

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Data scientist and software developer with more than six years of experience across local government, utilities, consulting, and marketing. My analytical work includes forecasting, machine learning, statistics, and geospatial analysis, along with developing dashboards, web applications, GIS tools, data pipelines, and automation.

Technology Analyst

City of Raleigh

Raleigh, NC

June 2025 - Present

- Build dashboards, automate recurring work, and investigate system and process problems for Planning and Development.
- Built a PyTorch computer vision application that detects and redacts engineering seals and other sensitive information in large PDF plan sets, reducing manual review.
- Produce city population estimates and long-range forecasts using census data, housing inventories, permit activity, and spatial modeling.
- Developed an ArcGIS Pro Python toolbox for internal use that downloads Census ACS 5-year tables, lists variable definitions, and creates feature classes with TIGERweb geometry for states, counties, places, tracts, and block groups.
- Built public dashboards and more than 20 internal Power BI dashboards on review timelines, permitting, inspections, annexations, staff workload, and development activity.
- Analyzed suburban and urban growth patterns and presented the findings to City Council and senior leadership.
- Automated meeting tracking, event management, and customer intake, and prepared staff documentation and training materials for department dashboards and data tools.

Data Scientist

Tennessee Valley Authority

Chattanooga, TN (Remote)

August 2022 - June 2025

- Developed machine learning models, statistical analyses, data pipelines, and R Shiny applications for environmental, geochemical, and operational projects.
- Built applications for groundwater evaluation, geochemical modeling, and billing anomaly detection.
- Built document-search and analysis tools with large language models, embeddings, retrieval-augmented generation, and LangChain.

- Prototyped new analytical methods and applications in Python, R, SQL, AWS, Azure DevOps, and Dataiku.
- Helped assess Domino Data Lab, DataRobot, Dataiku, SAS, and AWS SageMaker for an enterprise MLOps platform, focusing on model monitoring, reproducibility, workflow management, and deployment.
- Developed and taught advanced R programming courses for technical and non-technical employees.

Data Analyst

Elder Research Inc.

Raleigh, NC

August 2021 - August 2022

- Built a Databricks machine learning pipeline for ETL, model training, and deployment that identified supplier slippage drivers for Church & Dwight and supported more than \$1 million in savings.
- Designed an AWS workflow for Chick-fil-A using Redshift, Athena, and Apache Airflow to automate reconciliation between inventory and sales forecasting systems.
- Developed classification, forecasting, recommendation, and data integration solutions for commercial clients using Python, R, SQL, and cloud platforms.
- Worked with technical and non-technical teams to define requirements, check data quality, document methods, and present findings.

Data Science Intern

Merkle Inc.

New York City, NY (Remote)

June 2021 - August 2021

- Developed customer segmentation and propensity models for targeted marketing campaigns using Python, Databricks, and statistical modeling.
- Built data pipelines and an audience overlap analysis for the 2021 Jaguar and Land Rover campaigns, improving targeting efficiency by 15%.

Forecasting Analyst

Genworth

Lynchburg, VA

January 2020 - March 2021

- Developed time-series models to forecast call volume and staffing needs using Python and PostgreSQL.
- Accounted for seasonality and day-of-week effects in forecasts for incoming and outgoing call volume.
- Updated forecasts during the COVID-19 pandemic and communicated results for staffing, scheduling, and operational planning.

Education

Bachelor of Science in Data Analytics

- Western Governor's University
- Graduation: Fall, 2021

Graduate Certificate in Statistics

- North Carolina State University
- Completion: Spring, 2024

Master of Science in Geospatial Information Science and Technology

- North Carolina State University
- Graduation: Spring, 2025

Skills

Programming Languages

- Python
- R
- SAS
- SQL/NoSQL
- JavaScript/HTML/CSS

Software/Technologies

- Git
- Unix/Linux
- AWS (Athena, Redshift, EC2, Lambda)
- Dashboards (Power BI, Shiny, Dash)
- GIS Software (Esri Products, QGIS, GRASS GIS)
- MLOps Software (Dataiku, DataRobot, MLflow, Domino Data Lab)

Other Skills/Experience

- ETL and Data Integration
- Statistics & Applied Mathematics
- Machine Learning
- Deep Learning (PyTorch, TensorFlow, Keras)
- Natural Language Processing (NLP)
- LLMs (Retrieval-Augmented Generation, LangChain)
- Forecasting
- Geospatial Analytics & Spatial Statistics

Certifications

- AWS Cloud Practitioner, July 2024 - *Amazon Web Services (AWS)*